

Discrete Fourier Transform

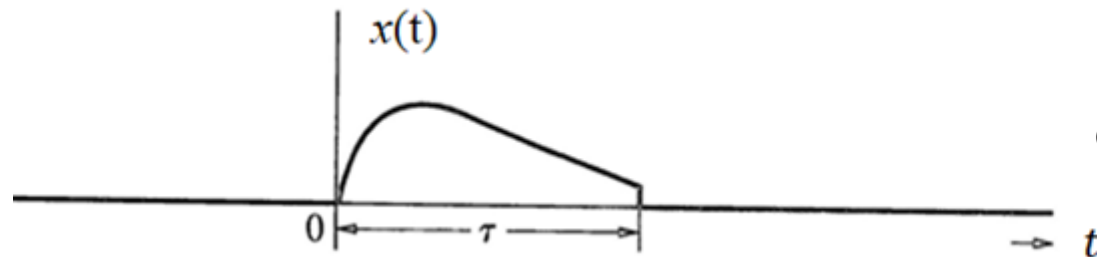
Chapter 9

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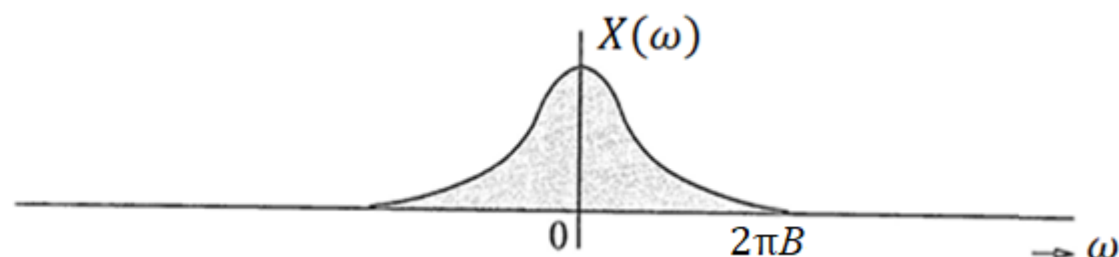
Outline

- The Discrete Fourier Transform
- Discrete-Time Filtering Using the DFT

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

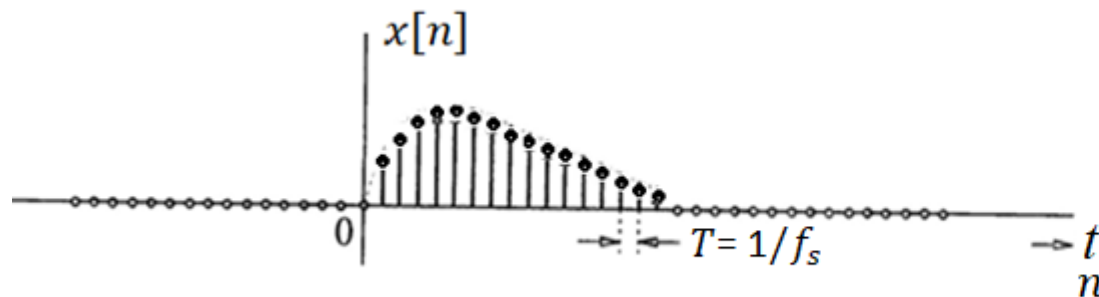


$\longleftrightarrow$
CTFT

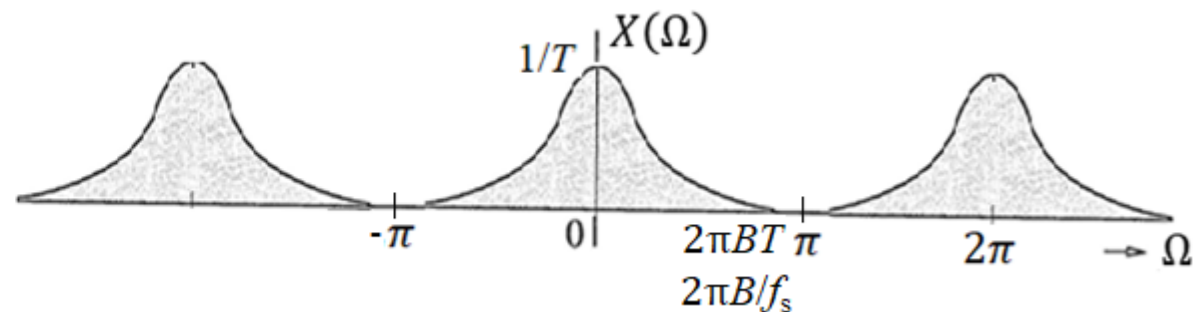


$$X(\Omega) = \sum_{n=-\infty}^{n=\infty} x[n] e^{-j\Omega n}$$

$$\Omega = \omega T = \omega / f_s$$

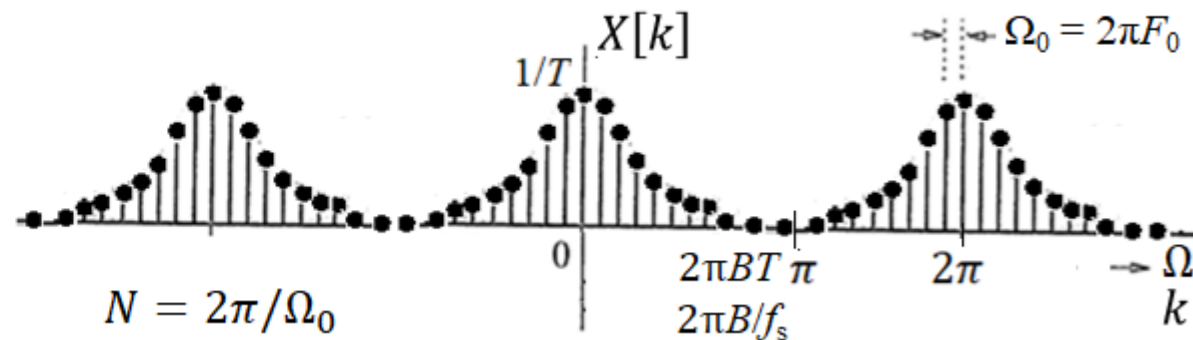
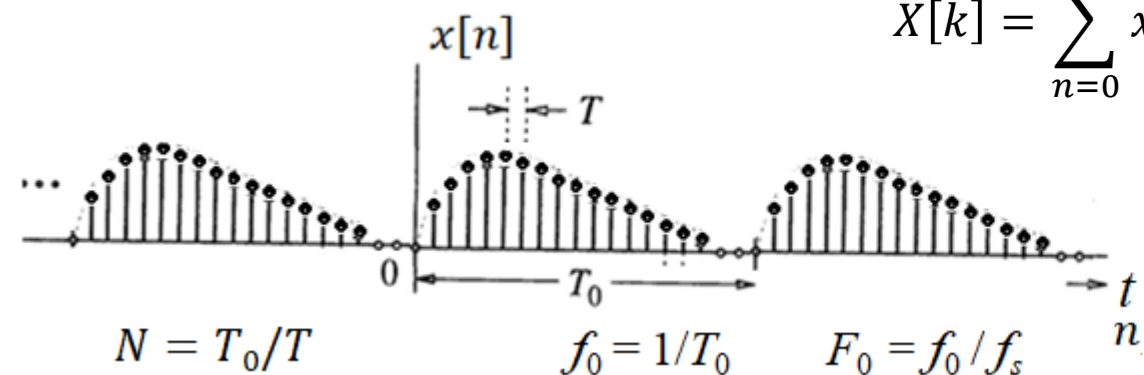


$\longleftrightarrow$
DTFT



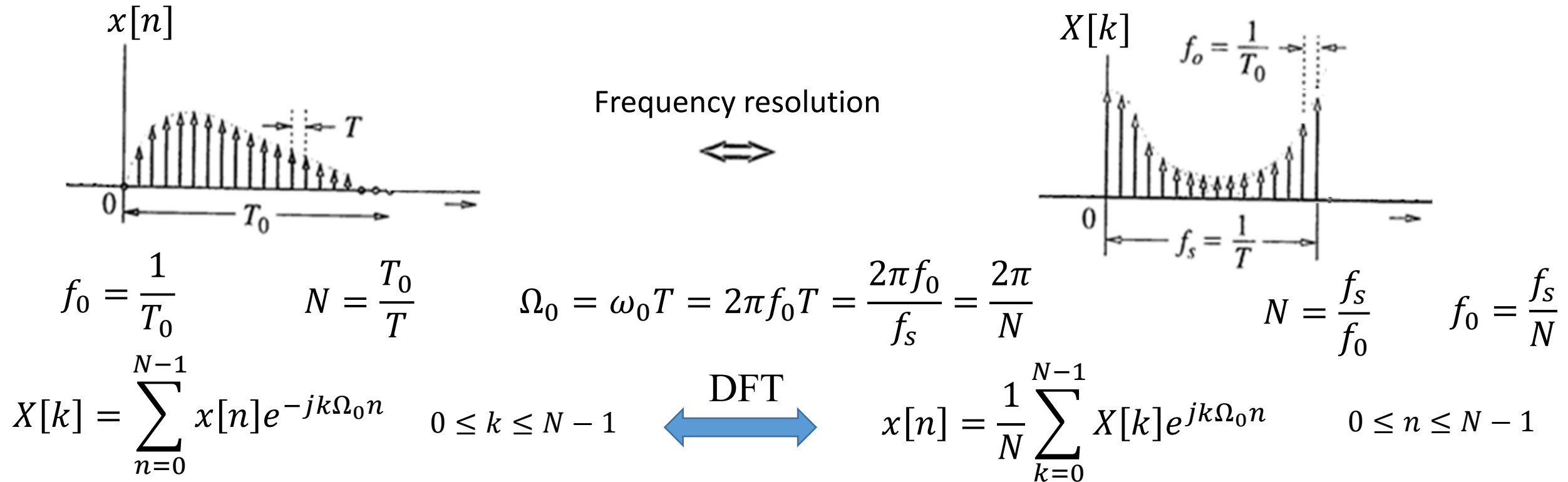
$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-jk\Omega_0 n} \quad 0 \leq k \leq N-1$$

$\longleftrightarrow$
DFT



The Discrete Fourier Transform

The direct and inverse relationships between a length- N signal $x[n]$ and the N samples of the spectrum $X[k]$ define the discrete Fourier transform (DFT). The DFT proves extremely useful in numerical computations of the discrete time Fourier transform and its inverse (DTFT).



$X[k] = X^*[N - k]$ where $k = 1, 2, \dots, (N-1)/2$ for odd N or $(N-2)/2$ for even N .

Example

Find the DFT of the length-3 signal $x[n] = 3\delta[n] + 2\delta[n-1] + 3\delta[n-2]$. Compare the DFT spectrum $X[k]$ with the corresponding DTFT spectrum $X(\Omega)$.

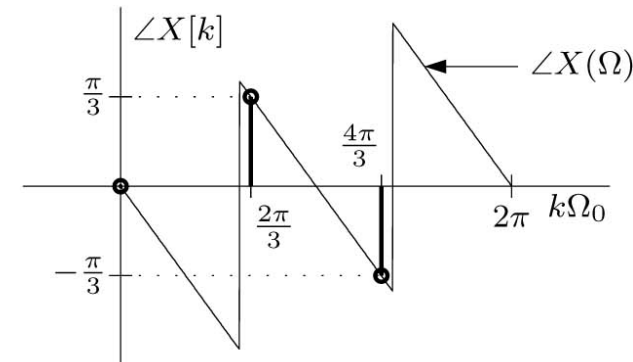
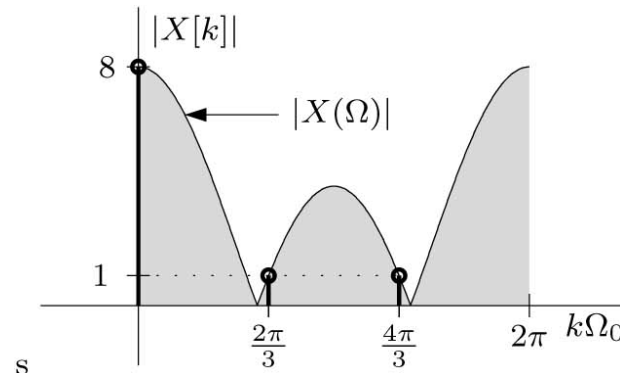
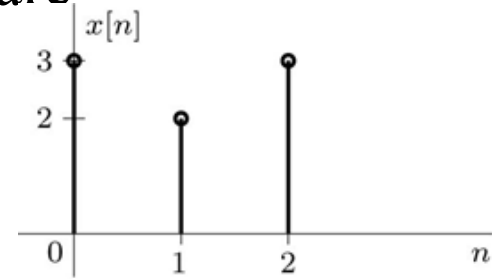
Solution $\Omega_0 = \frac{2\pi}{N} = \frac{2\pi}{3}$

$$X[k] = \sum_{n=0}^2 x[n]e^{-j\left(\frac{2\pi}{3}\right)kn} = 3 + 2e^{-j\frac{2\pi}{3}k} + 3e^{-j\frac{4\pi}{3}k} = e^{-j\frac{2\pi}{3}k} \left[2 + 6\cos\left(\frac{2\pi}{3}k\right) \right]$$

$$k = 0, 1, 2$$

$$X(\Omega) = \sum_{n=0}^2 x[n]e^{-j\Omega n} = e^{-j\Omega} [2 + 6\cos(\Omega)]$$

k	0	1	2
$ X[k] $	8	1	1
$\angle X[k]$	0	$\frac{\pi}{3}$	$-\frac{\pi}{3}$



The Discrete Fourier Transform with Zero Padding

The DFT yields samples of the DTFT spaced at frequency intervals of $\Omega_0 = 2\pi/N$. In this way, the DFT size N determines the frequency resolution Ω_0 . To see more frequency components of DTFT we improve the frequency resolution by increasing N (number of samples in the signal $x[n]$). Since the number of samples already determined based on the sampling theorem, we can increase N by padding $x[n]$ with zeros.

Zero padding allows us to obtain more samples of $X(\Omega)$ for clarity. Furthermore, this process does not change the underlying signal $x[n]$, which means that zero padding does not affect the underlying spectrum $X(\Omega)$.

Example: a signal $x(t) = \sin(2\pi t) + \sin(2\pi 1.5t) + \sin(2\pi 3t)$ sampled at the Nyquist rate $f_s = 6$ samples/second. The signal is sampled for 1 second. What are the DFT components $X[k]$? Will you be able to see the 1.5 Hz sinusoidal signal?

Example

Using zero padding, find the 6-point DFT of the length-3 signal $x[n] = 3\delta[n] + 2\delta[n-1] + 3\delta[n-2]$. Compare the 6-point DFT $X[k]$ with the corresponding DTFT spectrum $X(\Omega)$.

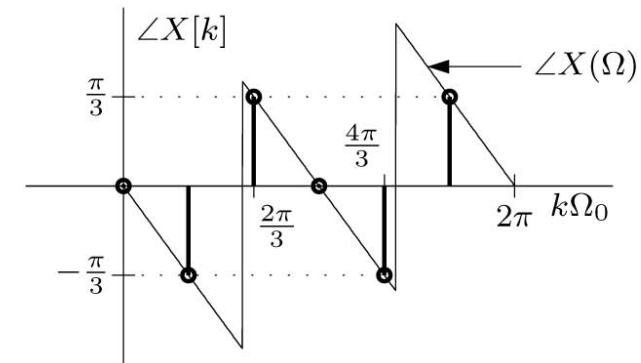
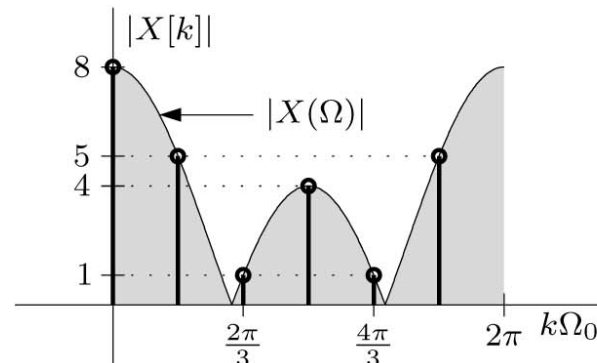
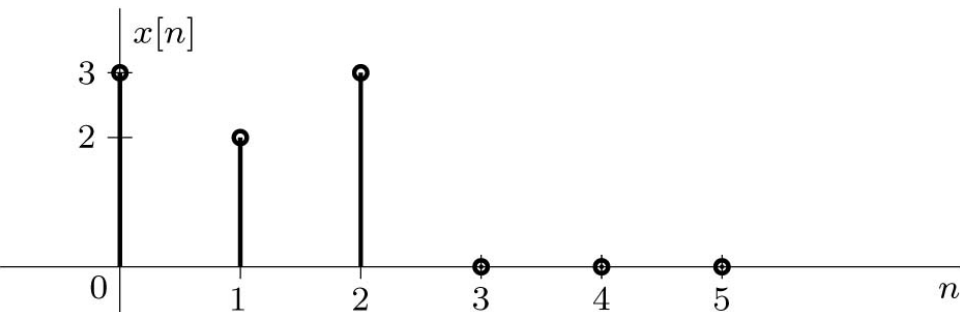
Solution

$$\Omega_0 = \frac{2\pi}{N} = \frac{2\pi}{6}$$

$$X[k] = \sum_{n=0}^5 x[n] e^{-j\left(\frac{2\pi}{6}\right)kn}$$

$$X(\Omega) = \sum_{n=0}^5 x[n] e^{-j\Omega n} = e^{-j\Omega} [2 + 6 \cos(\Omega)]$$

k	0	1	2	3	4	5
$ X[k] $	8	5	1	4	1	5
$\angle X[k]$	0	$-\frac{\pi}{3}$	$\frac{\pi}{3}$	0	$-\frac{\pi}{3}$	$\frac{\pi}{3}$



Matrix Representation of the DFT and Its Inverse

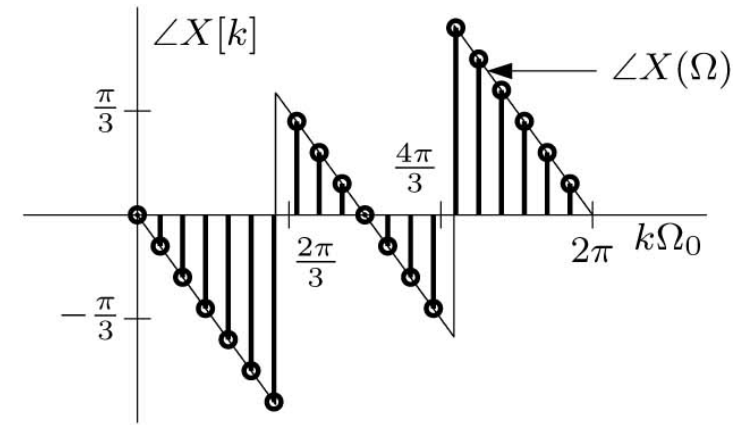
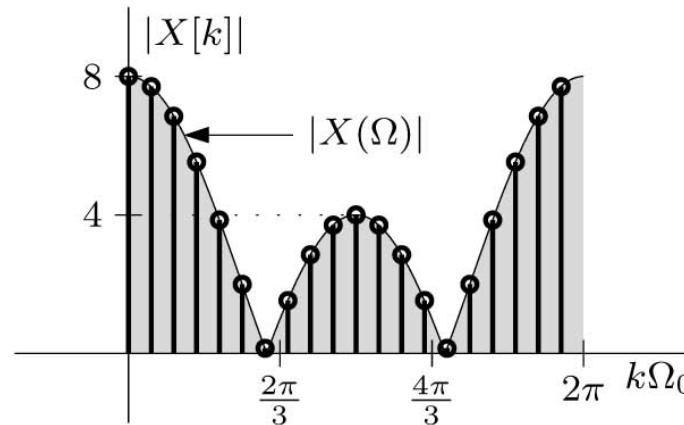
For large N it is cumbersome to manually evaluate the DFT and its inverse. Fortunately, both represent sets of N linear equations in N unknowns. Consequently, both can be expressed in matrix form, suitable for simple computer evaluation.

$$\begin{aligned}
 X[k] &= \sum_{n=0}^{N-1} x[n] e^{-j\Omega_0 kn} \\
 \underbrace{\begin{bmatrix} X[0] \\ X[1] \\ X[2] \\ \vdots \\ X[N-1] \end{bmatrix}}_{\mathbf{X}} &= \underbrace{\begin{bmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & e^{-j\Omega_0} & e^{-j\Omega_0 2} & \cdots & e^{-j\Omega_0(N-1)} \\ 1 & e^{-j\Omega_0 2} & e^{-j\Omega_0 4} & \cdots & e^{-j\Omega_0 2(N-1)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & e^{-j\Omega_0(N-1)} & e^{-j\Omega_0 2(N-1)} & \cdots & e^{-j\Omega_0(N-1)^2} \end{bmatrix}}_{\mathbf{D}_N} \underbrace{\begin{bmatrix} x[0] \\ x[1] \\ x[2] \\ \vdots \\ x[N-1] \end{bmatrix}}_{\mathbf{x}} \\
 x[n] &= \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j\Omega_0 kn} \\
 \underbrace{\begin{bmatrix} x[0] \\ x[1] \\ x[2] \\ \vdots \\ x[N-1] \end{bmatrix}}_{\mathbf{x}} &= \underbrace{\frac{1}{N} \begin{bmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & e^{j\Omega_0} & e^{j\Omega_0 2} & \cdots & e^{j\Omega_0(N-1)} \\ 1 & e^{j\Omega_0 2} & e^{j\Omega_0 4} & \cdots & e^{j\Omega_0 2(N-1)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & e^{j\Omega_0(N-1)} & e^{j\Omega_0 2(N-1)} & \cdots & e^{j\Omega_0(N-1)^2} \end{bmatrix}}_{\mathbf{D}_N^{-1}} \underbrace{\begin{bmatrix} X[0] \\ X[1] \\ X[2] \\ \vdots \\ X[N-1] \end{bmatrix}}_{\mathbf{X}}
 \end{aligned}$$

Example (read in textbook)

Use MATLAB to compute and plot the DFT of the length-3 signal $x[n] = 3\delta[n] + 2\delta[n-1] + 3\delta[n-2]$. Determine the zero padding needed to achieve a DFT frequency resolution of $\Omega_0 = \pi/10$. Compare the DFT spectrum $X[k]$ with the corresponding DTFT spectrum $X(\Omega)$.

Solution

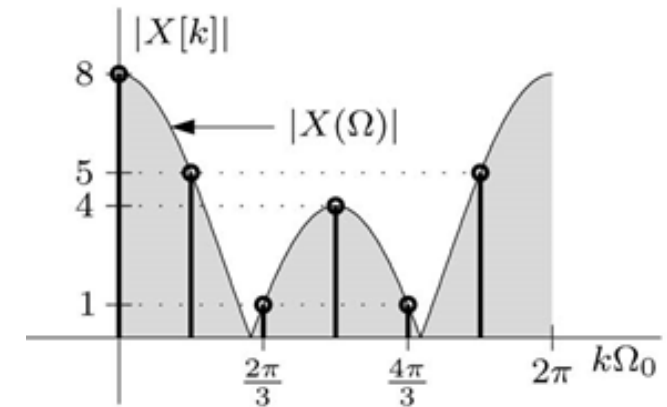


```
N = 20; Omega0 = 2*pi/N; k = (0:N-1); n = (0:N-1);  
x = [3;2;3;zeros(N-3,1)];  
DN = exp(-1j*Omega0*k'*n); X = DN*x;  
Omega = linspace(0,2*pi,6001); XOmega = exp(-1j*Omega).*(2+6*cos(Omega));  
subplot(121); stem(k*Omega0,abs(X)); line(Omega,abs(XOmega));  
subplot(122); stem(k*Omega0,angle(X)); line(Omega,angle(XOmega));
```

DFT Interpolation to Obtain the DTFT

We can determine the entire $X(\Omega)$ from the samples of the DFT using interpolation. This is similar to finding $x(t)$ from $x[n]$ samples by interpolation.

$$\begin{aligned} X(\Omega) &= \sum_{n=0}^{N-1} x[n] e^{-j\Omega n} = \sum_{n=0}^{N-1} \left[\frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j\Omega_0 k n} \right] e^{-j\Omega n} \\ &= \frac{1}{N} \sum_{k=0}^{N-1} X[k] \sum_{n=0}^{N-1} e^{-j(\Omega - \Omega_0 k) n} \end{aligned}$$



The inner sum on the right-hand side is a geometric progression and can be simplified as follow in the next slide.

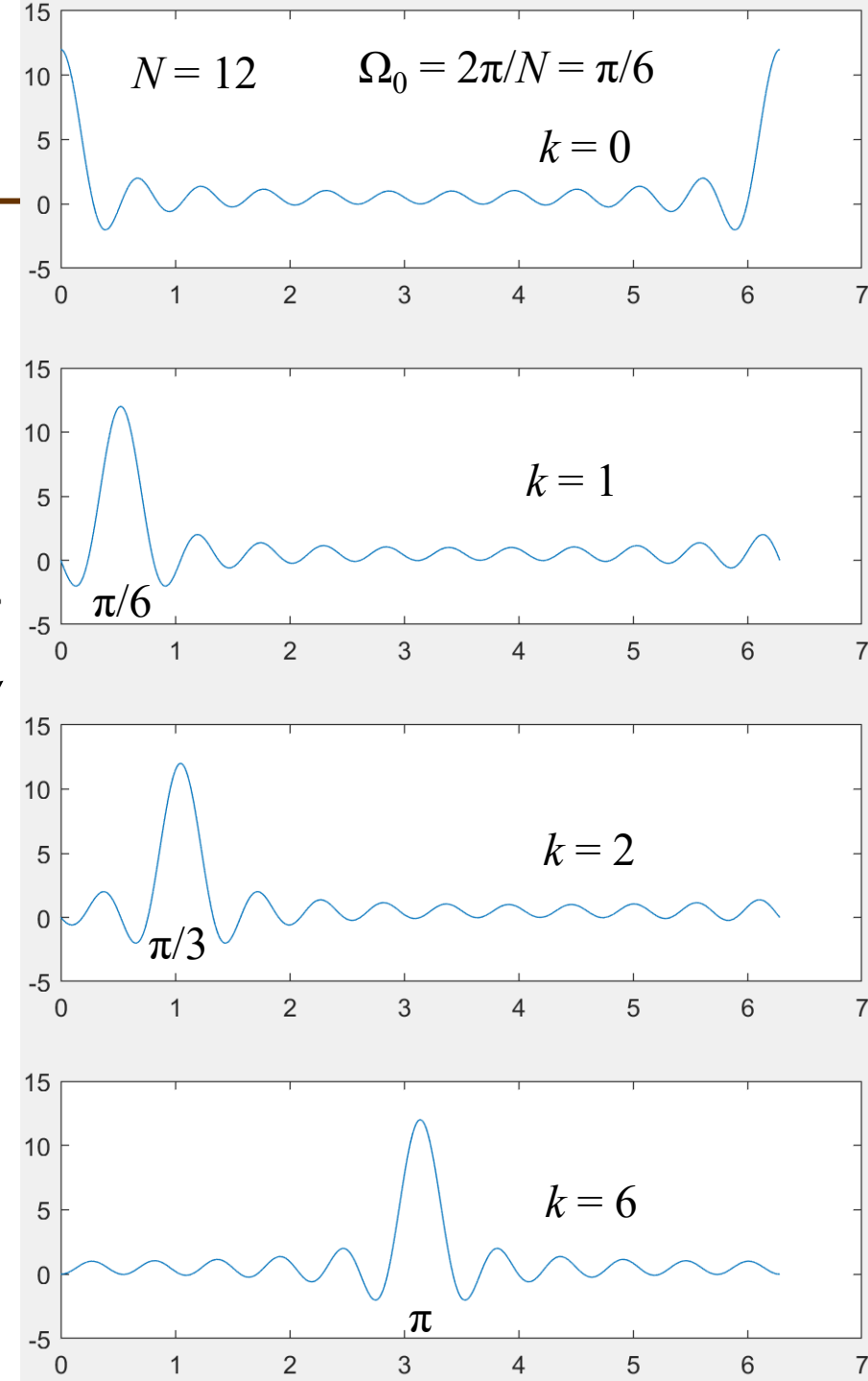
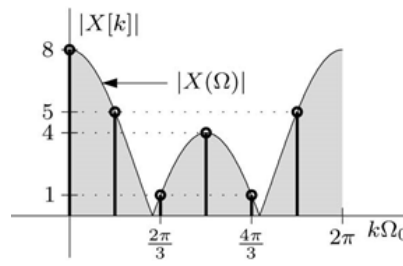
DFT Interpolation to Obtain the DTFT

$$\sum_{n=0}^{N-1} e^{-j(\Omega - \Omega_0 k)n} = \frac{1 - e^{-j(\Omega N - 2\pi k)}}{1 - e^{-j(\Omega - 2\pi k/N)}}$$

$$= \frac{(e^{j(\Omega N - 2\pi k)/2} - e^{-j(\Omega N - 2\pi k)/2}) e^{-j(\Omega N - 2\pi k)/2}}{(e^{j(\Omega N - 2\pi k)/2N} - e^{-j(\Omega N - 2\pi k)/2N}) e^{-j(\Omega N - 2\pi k)/2N}}$$

$$= \frac{\sin\left(\frac{\Omega N - 2\pi k}{2}\right)}{\sin\left(\frac{\Omega N - 2\pi k}{2N}\right)} e^{-j(\Omega N - 2\pi k)(N-1)/2N}$$

$$X(\Omega) = \frac{1}{N} \sum_{k=0}^{N-1} X[k] \frac{\sin\left(\frac{\Omega N - 2\pi k}{2}\right)}{\sin\left(\frac{\Omega N - 2\pi k}{2N}\right)} e^{-j(\Omega N - 2\pi k)(N-1)/2N}$$



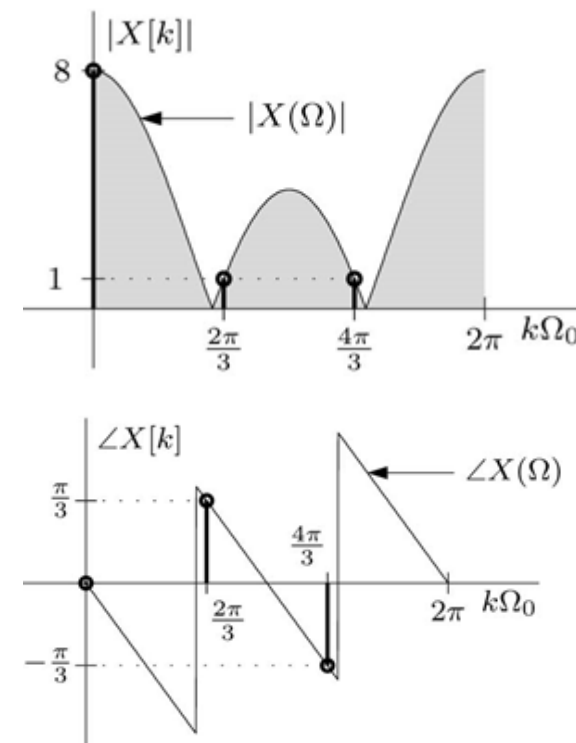
Example (read it textbook)

Using interpolation, determine the DTFT $X(\Omega)$ if the DFT of the corresponding 3-point $x[n]$ is given by $X[0] = 8$, $X[1] = e^{j\pi/3}$, and $X[2] = e^{-j\pi/3}$.

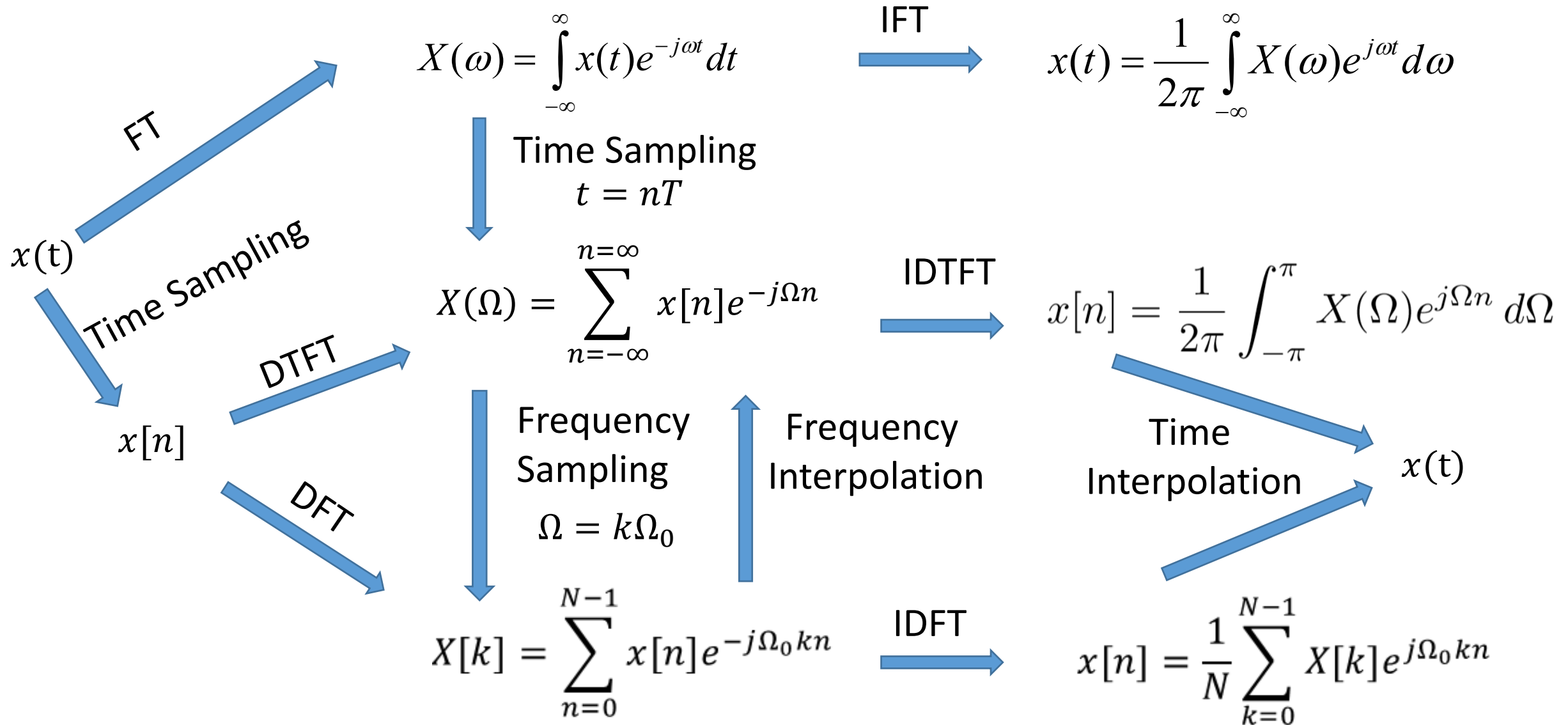
Solution

$$\begin{aligned} X(\Omega) &= \frac{1}{3} \sum_{k=0}^2 X[k] \frac{\sin\left(\frac{3\Omega - 2\pi k}{2}\right)}{\sin\left(\frac{3\Omega - 2\pi k}{6}\right)} e^{-j\left(\Omega - \frac{2\pi k}{3}\right)} \\ &= e^{-j\Omega} \frac{\sin(3\Omega/2)}{3} \left(\frac{8}{\sin(\Omega/2)} + \frac{1}{\sin(\frac{\Omega}{2} - \frac{\pi}{3})} - \frac{1}{\sin(\frac{\Omega}{2} - \frac{2\pi}{3})} \right) \end{aligned}$$

$$X(\Omega) = e^{-j\Omega} [2 + 6 \cos(\Omega)]$$



DTFT $\leftrightarrow$ DFT



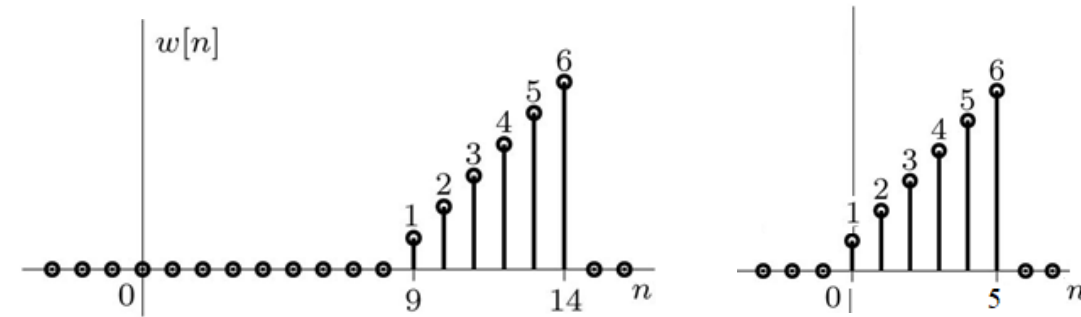
Confining $x[n]$ to $0 \leq n \leq N-1$ for Unique DFT

The summation in the DFT definition starts from $n = 0$ to $N-1$, so how do we find the DFT for a signal that starts at $n = 9$?

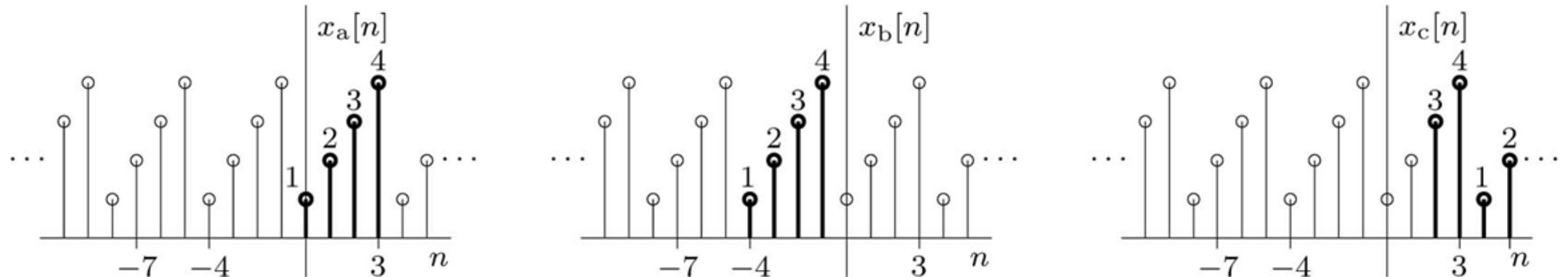
$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j\Omega_0 kn}$$

Can we just shift it to start at zero? **NO**

So, how do we confine $x[n]$ to $0 \leq n \leq N-1$?



The DFT for the signals $x_a[n]$, $x_b[n]$, and $x_c[n]$, are the same.

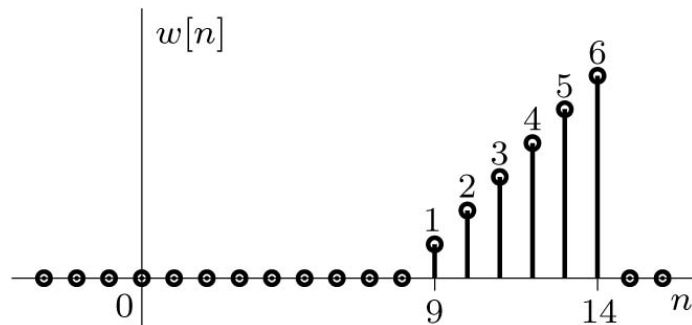


How to Confine $x[n]$ to $0 \leq n \leq N-1$ for Unique DFT

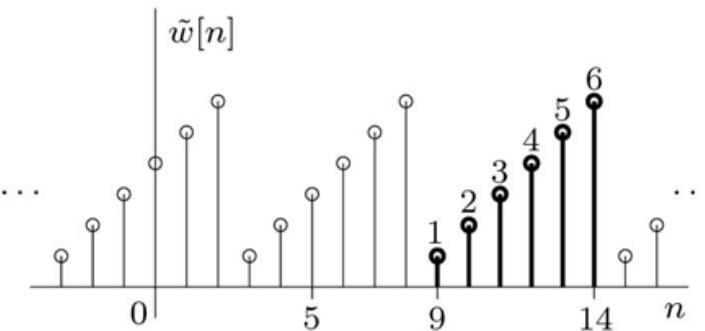
The DFT for the signals $w[n]$ and $x[n]$ of size $N = 6$ are the same.

$$W[k] = \sum_{n=9}^{14} x[n] e^{-j\frac{2\pi}{6}kn}$$

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j\Omega_0 kn}$$



The inverse DFT will produce a periodic $x[n]$ signal.

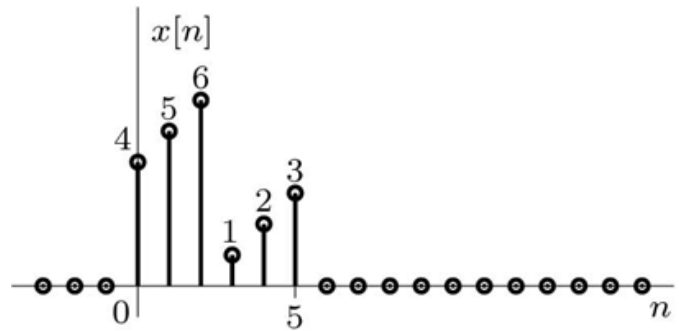


$n = 9$ to 14

$$W[k] = 1e^{-j\frac{2\pi}{6}9k} + 2e^{-j\frac{2\pi}{6}10k} + 3e^{-j\frac{2\pi}{6}11k} + 4e^{-j\frac{2\pi}{6}12k} + \dots + 6e^{-j\frac{2\pi}{6}14k}$$

$n = 0$ to 5

$$X[k] = 4 + 5e^{-j\frac{2\pi}{6}k} + 6e^{-j\frac{2\pi}{6}2k} + 1e^{-j\frac{2\pi}{6}3k} + \dots + 3e^{-j\frac{2\pi}{6}5k}$$



Confining $x[n]$ to $0 \leq n \leq N-1$ for Unique DFT

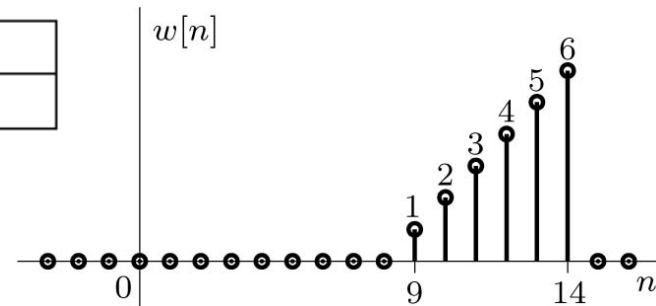
Modulo- N Shift

- Modulo- N is used in DFT to shift an arbitrary N -point sequence into the range 0 to $N-1$.
- The N uniform samples of the DTFT of a length- N sequence remain unchanged under the modulo- N shift of the sequence.
- Modulo- N value of a **positive** n is the remainder of the division of n by N .
- Modulo- N value of a **negative** n is N minus the remainder of the division of $|n|$ by N .

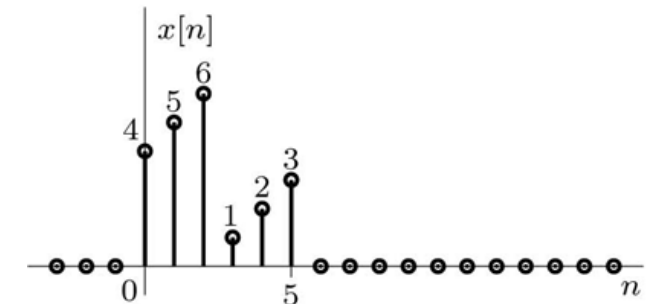
n	9	10	11	12	13	14
$\langle n \rangle_6$	3	4	5	0	1	2

n	-9	-10	-11	-12	-13	-14
$\langle n \rangle_6$	3	2	1	0	5	4

$$\langle n \rangle_N = n - rN \quad \text{where} \quad 0 \leq n - rN \leq N - 1 \quad r = \pm 1, 2, 3, \dots$$



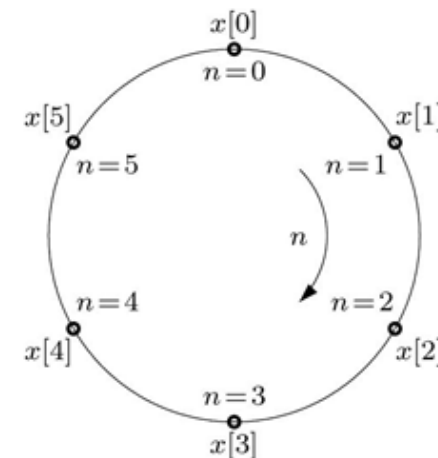
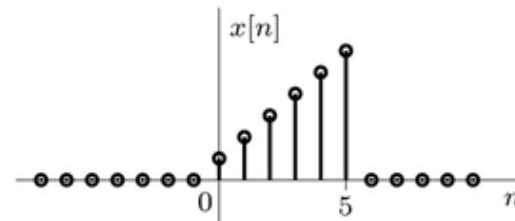
If DFT requires the confinement of the signal $x[n]$ between 0 and $N-1$ then how do we perform time shifting, reversal, and convolution for signal processing?



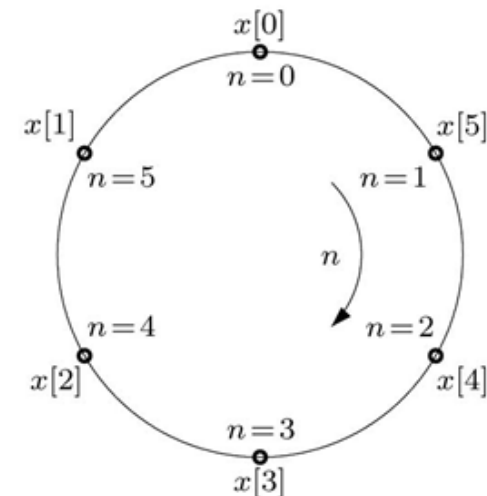
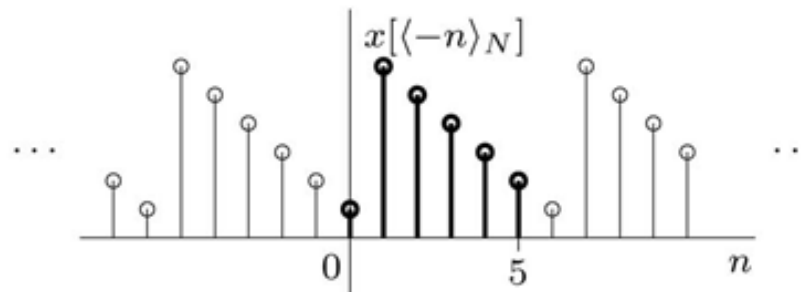
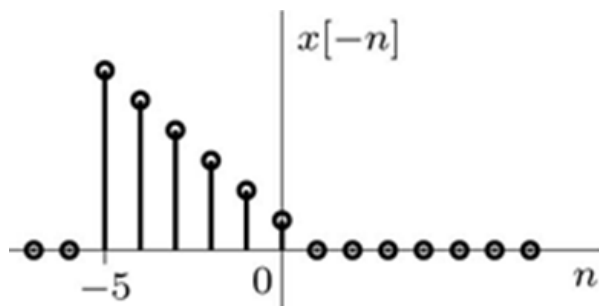
Modulo-N Shift of $x[-n]$

The Modulo-N Reflection

$$x[\langle -n \rangle_N] = \begin{cases} x[0] & n = 0 \\ x[N - n] & 1 \leq n \leq N - 1 \end{cases}$$



Just as with a linear representation, the circular representation of time reversal is just a reflection about the vertical axis.



Time Reversal Property

$$\text{if } x[n] \longleftrightarrow X[k], \text{ then } x[\langle -n \rangle_N] \longleftrightarrow X[\langle -k \rangle_N] = X[N - k]$$

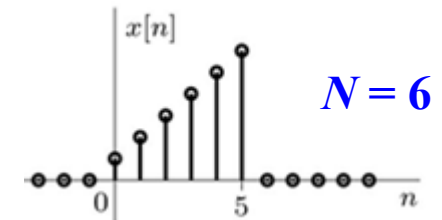
Modulo- N Shift of $x[n-m]$

The Circular Shift Modulo- N Reflection $x[\langle n - m \rangle_N]$

A linear shift of $x[n]$ by m samples yields the signal $x[n - m]$, which is a length- N signal with a range $m \leq n \leq m + N - 1$.

As with the case of reflection, we instead seek a form that will retain the range $0 \leq n \leq N - 1$. A modulo- N shift of $x[n - m]$, also called a circular shift, accomplishes exactly this task.

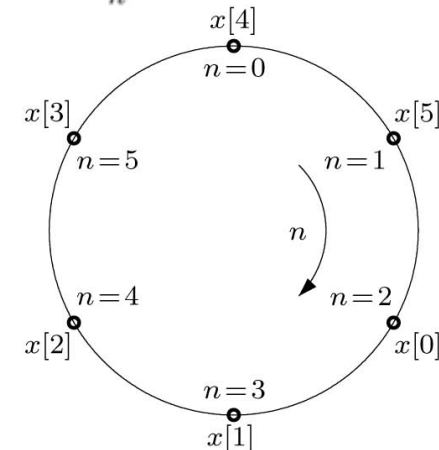
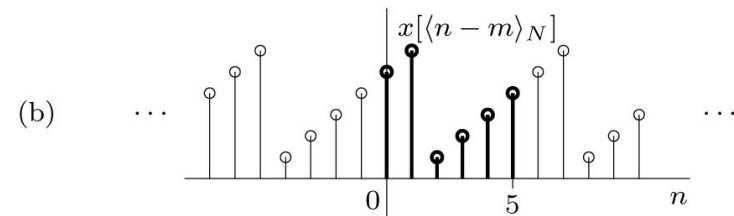
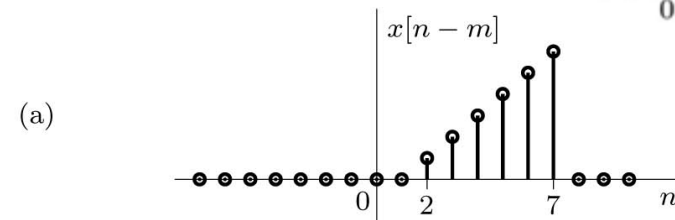
$$x[\langle n - m \rangle_N] = \begin{cases} x[n - m] & 0 \leq m \leq n \leq N - 1 \\ x[n - m + N] & 0 \leq n < m \leq N - 1 \end{cases}$$



Circular Shifting Property

if $x[n] \longleftrightarrow X[k]$, then

$$x[\langle n - m \rangle_N] \longleftrightarrow X[k]e^{-j\Omega_0 km}$$

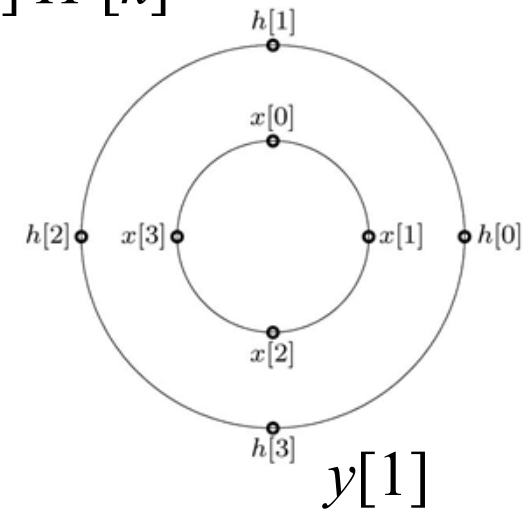
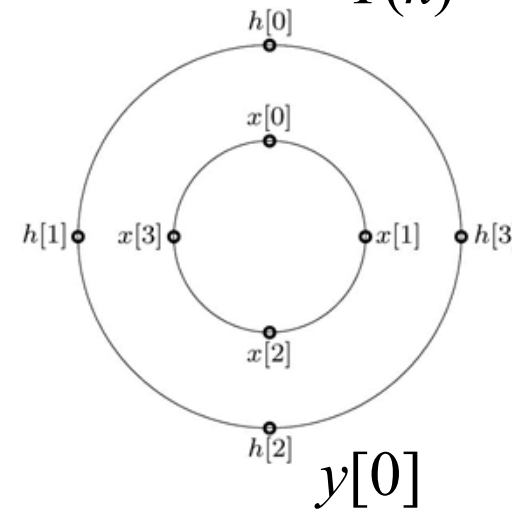


(c)

Circular Convolution

$$y[n] = x[n] \circledast h[n] = \sum_{m=0}^{N-1} x[m] h[\langle n - m \rangle_N] \longleftrightarrow Y(k) = X[k] H[k]$$

The two sequences to be convolved must have the same length. If not, we should append enough zeros to the smaller sequence to ensure equal lengths of both sequences.

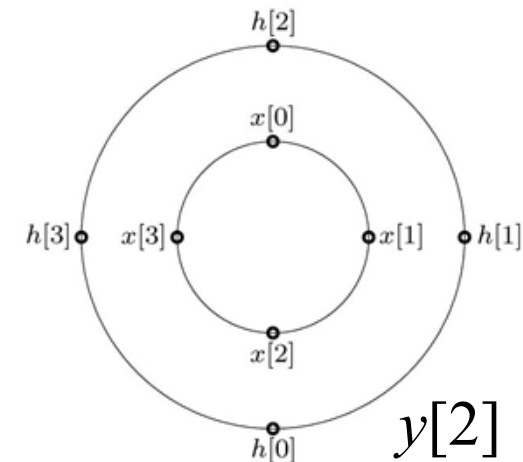
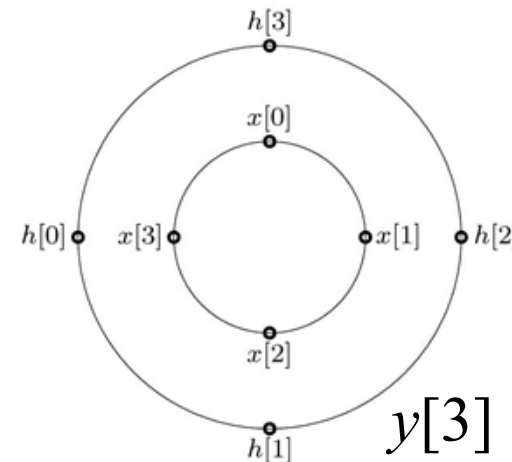


$$y[0] = x[0]h[0] + x[1]h[3] + x[2]h[2] + x[3]h[1]$$

$$y[1] = x[0]h[1] + x[1]h[0] + x[2]h[3] + x[3]h[2]$$

$$y[2] = x[0]h[2] + x[1]h[1] + x[2]h[0] + x[3]h[3]$$

$$y[3] = x[0]h[3] + x[1]h[2] + x[2]h[1] + x[3]h[0]$$



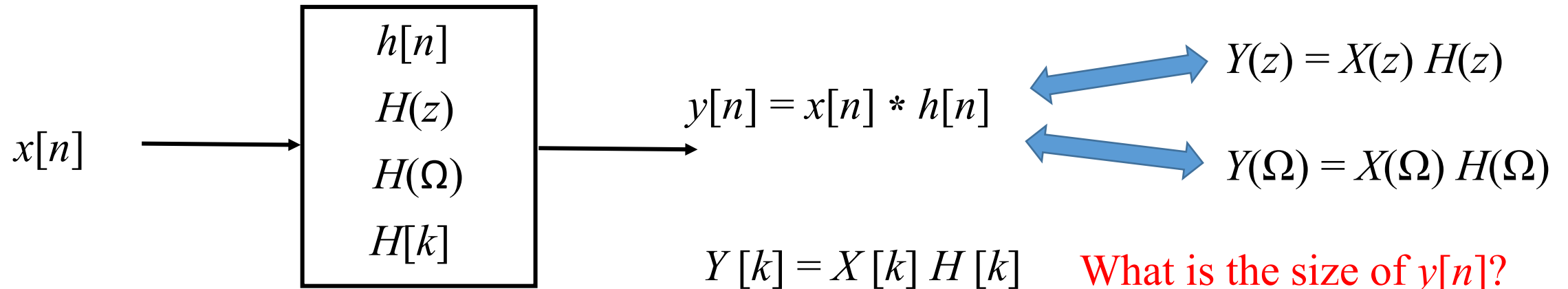
Discrete Fourier Transform	Discrete-Time Fourier Transform
Synthesis: $x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j\Omega_0 kn}, \quad \Omega_0 = \frac{2\pi}{N}$ Analysis: $X[k] = \sum_{n=0}^{N-1} x[n] e^{-j\Omega_0 kn}, \quad \Omega_0 = \frac{2\pi}{N}$ Duality: if $x[n] \longleftrightarrow X[k]$, then $X[n] \longleftrightarrow Nx[\langle -k \rangle_N]$ Linearity: $ax[n] + by[n] \longleftrightarrow aX[k] + bY[k]$ Complex Conjugation: $x^*[n] \longleftrightarrow X^*[\langle -k \rangle_N]$ $x^*[\langle -n \rangle_N] \longleftrightarrow X^*[k]$ Reversal: $x[\langle -n \rangle_N] \longleftrightarrow X[\langle -k \rangle_N]$ Shifting: $x[\langle n - m \rangle_N] \longleftrightarrow X[k] e^{-j\Omega_0 km}, \quad \Omega_0 = \frac{2\pi}{N}$ $x[n] e^{j\Omega_0 mn} \longleftrightarrow X[\langle k - m \rangle_N], \quad \Omega_0 = \frac{2\pi}{N}$ Convolution: $x[n] \circledast y[n] \longleftrightarrow X[k] Y[k]$ $x[n] y[n] \longleftrightarrow \frac{1}{N} X[k] \circledast Y[k]$ Correlation: $\rho_{x,y}[l] = x[l] \circledast y^*[\langle -l \rangle_N] \longleftrightarrow X[k] Y^*[k]$ Parseval's: $E_x = \sum_{n=0}^{N-1} x[n] ^2 = \frac{1}{N} \sum_{k=0}^{N-1} X[k] ^2$	Synthesis: $x[n] = \frac{1}{2\pi} \int_{2\pi} X(\Omega) e^{j\Omega n} d\Omega$ Analysis: $X(\Omega) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\Omega n}$ Duality: Linearity: $ax[n] + by[n] \Longleftrightarrow aX(\Omega) + bY(\Omega)$ Complex Conjugation: $x^*[n] \Longleftrightarrow X^*(-\Omega)$ $x^*[-n] \Longleftrightarrow X^*(\Omega)$ Reversal: $x[-n] \Longleftrightarrow X(-\Omega)$ Shifting: $x[n - m] \Longleftrightarrow X(\Omega) e^{-j\Omega m}$ $x[n] e^{j\Omega_0 n} \Longleftrightarrow X(\Omega - \Omega_0)$ Convolution: $x[n] * y[n] \Longleftrightarrow X(\Omega) Y(\Omega)$ $x[n] y[n] \Longleftrightarrow \frac{1}{2\pi} X(\Omega) \circledast Y(\Omega)$ Correlation: $\rho_{x,y}[l] = x[l] * y^*[-l] \Longleftrightarrow X(\Omega) Y^*(\Omega)$ Parseval's: $E_x = \sum_{n=-\infty}^{\infty} x[n] ^2 = \frac{1}{2\pi} \int_{2\pi} X(\Omega) ^2 d\Omega$

Signal Processing and Filtering Using the DFT

Find the output $y[n]$ of the system described by the impulse response

$h[n] = \delta[n] + \delta[n-1] + \delta[n-2]$ and with input $x[n] = \delta[n-1] + 2\delta[n-2] + 3\delta[n-3] + 4\delta[n-4]$

Size of $y[n]$: $N_y = N_x + N_h - 1$



Will the inverse DFT of $Y[k]$ give us $y[n]$?

Circular and Linear Convolution

- Circular Convolution is different from linear convolution. In Circular all N samples of $x[n]$ and $h[n]$ are aligned and multiplied for any value of n .
- For linear system analysis we need to modify circular convolution to be similar to linear convolution.
- The length N of the signal $y[n]$ which is the result of convolving $x[n]$ with $h[n]$ is $N = N_x + N_h - 1$.
- If linear and circular convolutions are to yield identical results, then a necessary condition is that the length of each signal $x[n]$ and $h[n]$ must be extended to $N = N_x + N_h - 1$. In other words, we must pad $N_h - 1$ zeros to $x[n]$ and pad $N_x - 1$ zeros to $h[n]$.

Discrete-Time Filtering Using the DFT

The DFT is useful in the computation of direct and inverse Fourier transforms, filtering, convolution, correlation, and others. It can perform these tasks not only for discrete-time systems but also for continuous-time signals and systems. The efficient FFT algorithm, makes the DFT even more appealing.

The DFT-based procedure to find the linear convolution of two signals $x[n]$ and $h[n]$, whose lengths are N_x and N_h , respectively, is summarized in four steps as follows:

1. Pad $N_h - 1$ zeros to $x[n]$ to form $x_{\text{pad}}[n]$, and pad $N_x - 1$ zeros to $h[n]$ to form $h_{\text{pad}}[n]$.
2. Find $X_{\text{pad}}[k]$ and $H_{\text{pad}}[k]$, the DFTs of $x_{\text{pad}}[n]$ and $h_{\text{pad}}[n]$.
3. Multiply $X_{\text{pad}}[k]$ by $H_{\text{pad}}[k]$ to obtain $Y[k]$.
4. The desired convolution $y[n]$ is the IDFT of $Y[k]$.

Example: $N_x = 7$ $N_h = 3$

$$\{Y[k]\} = \{32, 4\sqrt{2}e^{-j\frac{3\pi}{4}}, 0, 4\sqrt{2}e^{j\frac{3\pi}{4}}\} \quad \text{Example}$$

Using DFT-based convolution, determine the output $y[n]$ of an LTID filter with impulse response $h[n] = 2\delta[n] + 2\delta[n - 1]$ to an input $x[n] = 3\delta[n] + 2\delta[n - 1] + 3\delta[n - 2]$.

Solution

After padding $h[n]$ and $x[n]$ with zeros, $N = 4$ and $\Omega_0 = \pi/2$.

$$x_{\text{pad}}[n] = [3, 2, 3, 0]$$

$$h_{\text{pad}}[n] = [2, 2, 0, 0]$$

$$X_{\text{pad}}[k] = \sum_{n=0}^3 x_{\text{pad}}[n]e^{-j\Omega_0 kn} = \sum_{n=0}^3 x[n]e^{-j\frac{\pi}{2}kn} = 3 + 2e^{-j\frac{\pi}{2}k} + 3e^{-j\pi k} \quad 0 \leq k \leq 3$$

$$[8 \quad -j2 \quad 4 \quad j2]$$

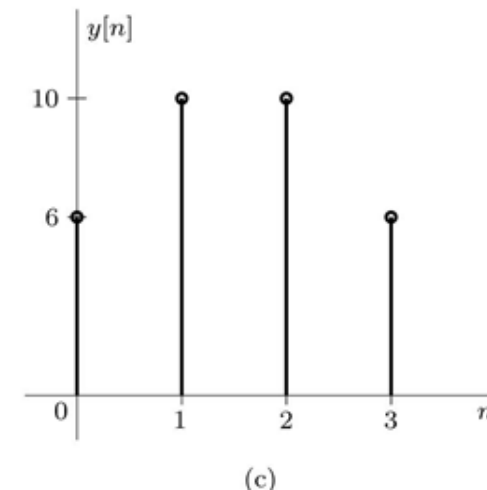
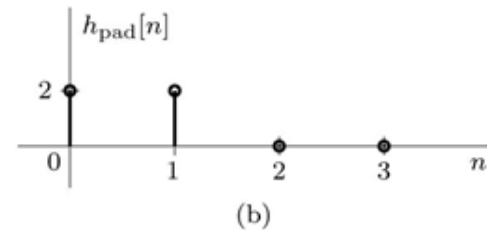
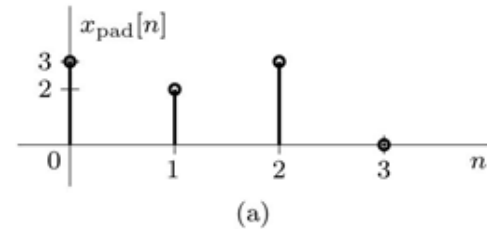
$$H_{\text{pad}}[k] = \sum_{n=0}^3 h_{\text{pad}}[n]e^{-j\Omega_0 kn} = \sum_{n=0}^1 h[n]e^{-j\frac{\pi}{2}kn} = 2 + 2e^{-j\frac{\pi}{2}k} \quad 0 \leq k \leq 3$$

$$[4 \quad 2-j2 \quad 0 \quad 2+j2]$$

$$Y[k] = 6 + 10e^{-j0.5\pi k} + 10e^{-j\pi k} + 6e^{-j1.5\pi k}$$

$$[32 \quad -4-j4 \quad 0 \quad -4+j4]$$

$$y[n] = \frac{1}{4} \sum_{k=0}^3 Y[k]e^{j\Omega_0 kn} = \frac{1}{4} \left(Y[0] + Y[1]e^{j\frac{\pi}{2}n} + Y[2]e^{j\pi n} + Y[3]e^{j\frac{3\pi}{2}n} \right)$$



Continue Example

Solution

$$y[n] = 8 + 4\sqrt{2}e^{-j\frac{3\pi}{4}+j\frac{\pi}{2}n} + 4\sqrt{2}e^{j\frac{3\pi}{4}-j\frac{\pi}{2}n} = 8 + 2\sqrt{2}\cos\left(\frac{\pi}{2}n - \frac{3\pi}{4}\right), \quad 0 \leq n \leq 3$$

$$\{y[n]\} = \{6, 10, 10, 6\}$$

For small signals, the convolution might be easier than DFT, but for large signals the DFT is faster

```
x = [3;2;3]; h = [2;2]; Nx = length(x); Nh = length(h);  
xpad = [x;zeros(Nh-1,1)]; hpad = [h;zeros(Nx-1,1)];  
N = Nx+Nh-1; n = (0:N-1); k=(0:N-1); Omega0 = 2*pi/N; DN = exp(-1j*Omega0*k'*n);  
Xpad = DN*xpad; Hpad = DN*hpad; y = conj(DN)/N*(Xpad.*Hpad)
```

Ans y = 6 10 10 6

Block Convolution

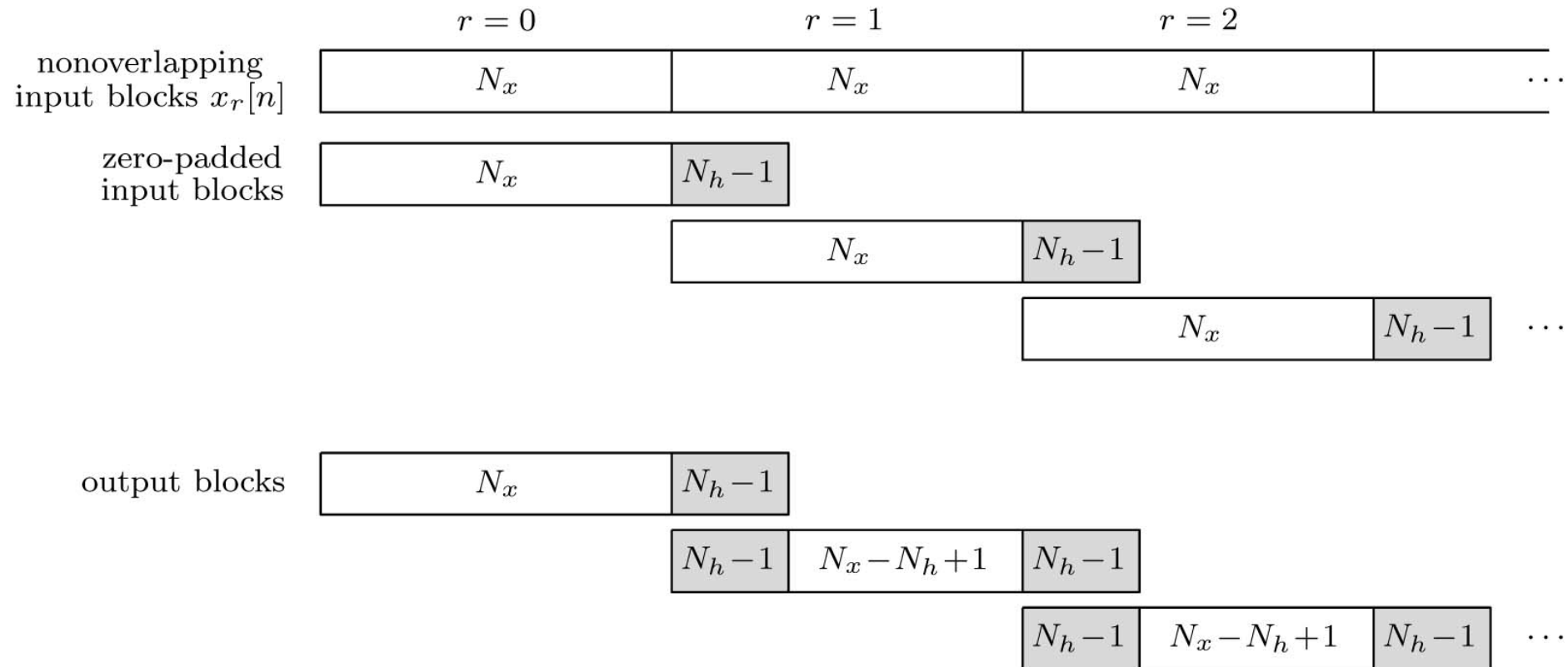
In real application the input signal is acquired in real time so its length is not known and it may be very large. This will cause delay and excessive use of memory and processing time. Block convolution is the solution. It break the coming input signal into blocks of determined length N and convolve each block with $h[n]$ and then add the results of all the convolutions.

There are two methods for block convolution:

- 1- Overlap-and-Add
- 2- Overlap-and-Save

Overlap-and-Add Method

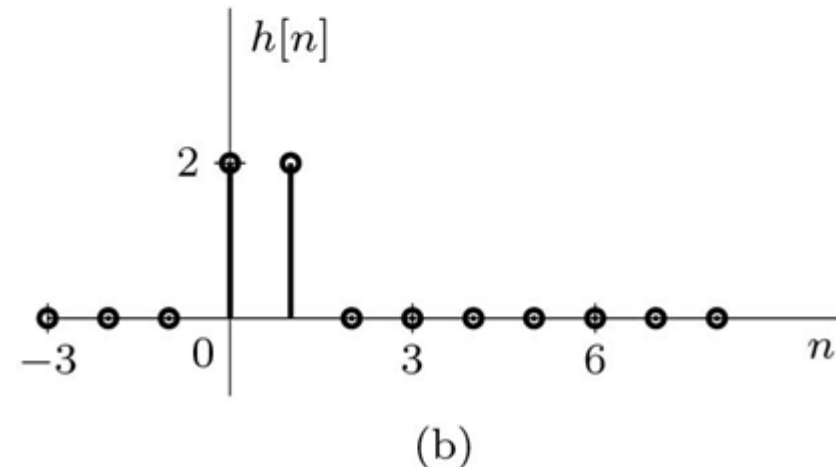
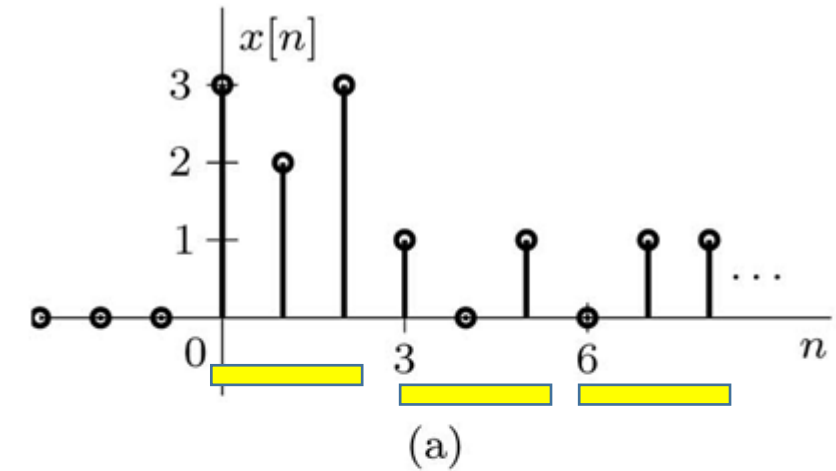
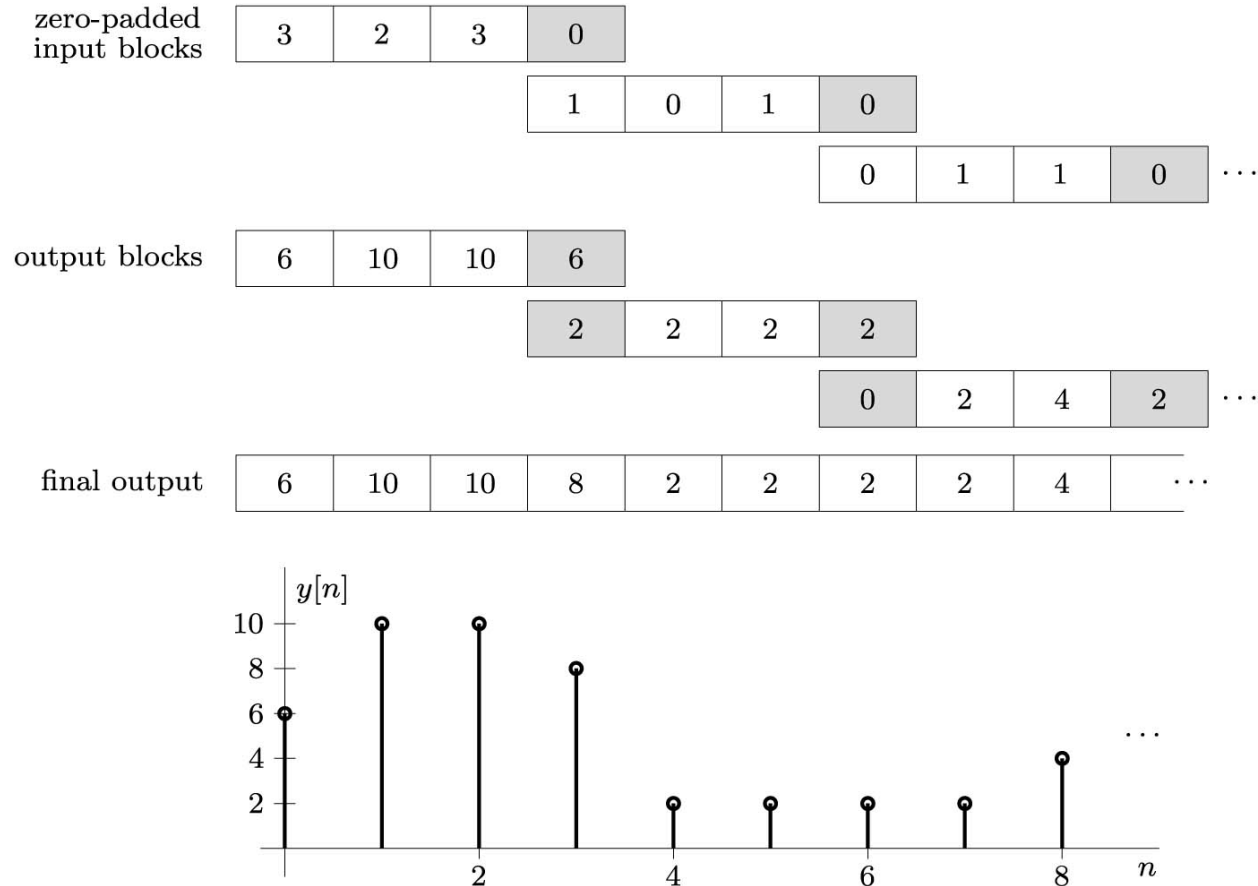
In practice the length of the filter impulse response $h[n]$, N_h , is usually smaller than the length of the input blocks N_x . For circular convolution we pad N_h-1 zeros at the end of each data block.



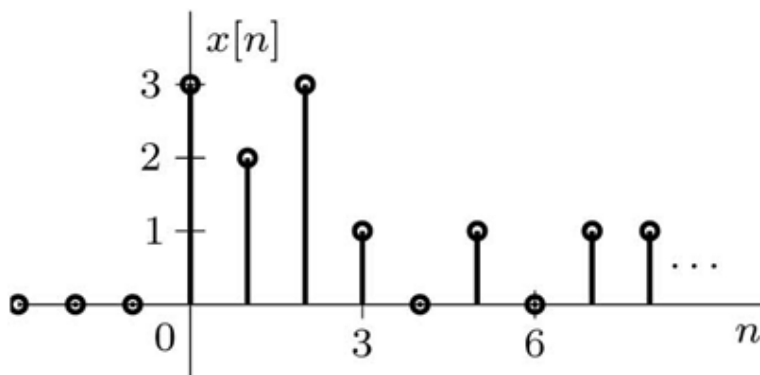
Example

Using the overlap-and-add method with an $N_x = 3$ input block size, find the response $y[n]$ of an LTID system with the $N_h = 2$ finite impulse response $h[n]$ and input $x[n]$ shown in Figure below.

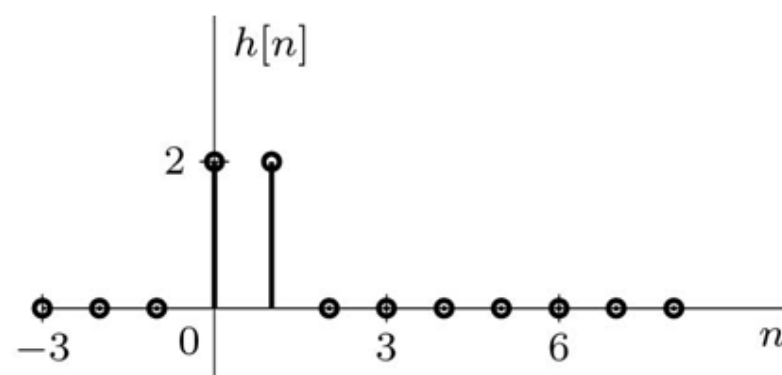
Solution



Continue Example



$$X_{r,\text{pad}}[k] = \sum_{n=0}^2 x[n + rN_x] e^{-j\frac{\pi}{2}kn}$$



$$H_{\text{pad}}[k] = \sum_{n=0}^1 h[n] e^{-j\frac{\pi}{2}kn}$$

$$X_{0,\text{pad}}[k] = 3 + 2e^{-j\frac{\pi}{2}k} + 3e^{-j\pi k}$$

$$H_{\text{pad}}[k] = 2 + 2e^{-j\frac{\pi}{2}k}$$

$$Y_0[k] = X_{0,\text{pad}}[k] H_{\text{pad}}[k]$$

$$y_0[n] = \frac{1}{4} \sum_{k=0}^3 Y_0[k] e^{j\frac{\pi}{2}kn}$$

Continue Example (Matlab Code)

```
Nx = 3; Nh = 2; N = Nx+Nh-1; Omega0 = 2*pi/N; k = (0:N-1); n = (0:N-1);  
DN = exp(-1j*Omega0*k'*n); x0pad = [3;2;3;zeros(Nh-1,1)]; hpad = [2;2;zeros(Nx-1,1)];  
X0pad = DN*x0pad; Hpad = DN*hpad; y0 = conj(DN)/N*(X0pad.*Hpad)
```

Ans y0 = 6 10 10 6

```
x1pad = [1;0;1;zeros(Nh-1,1)]; X1pad = DN*x1pad; y1 = conj(DN)/N*(X1pad.*Hpad)
```

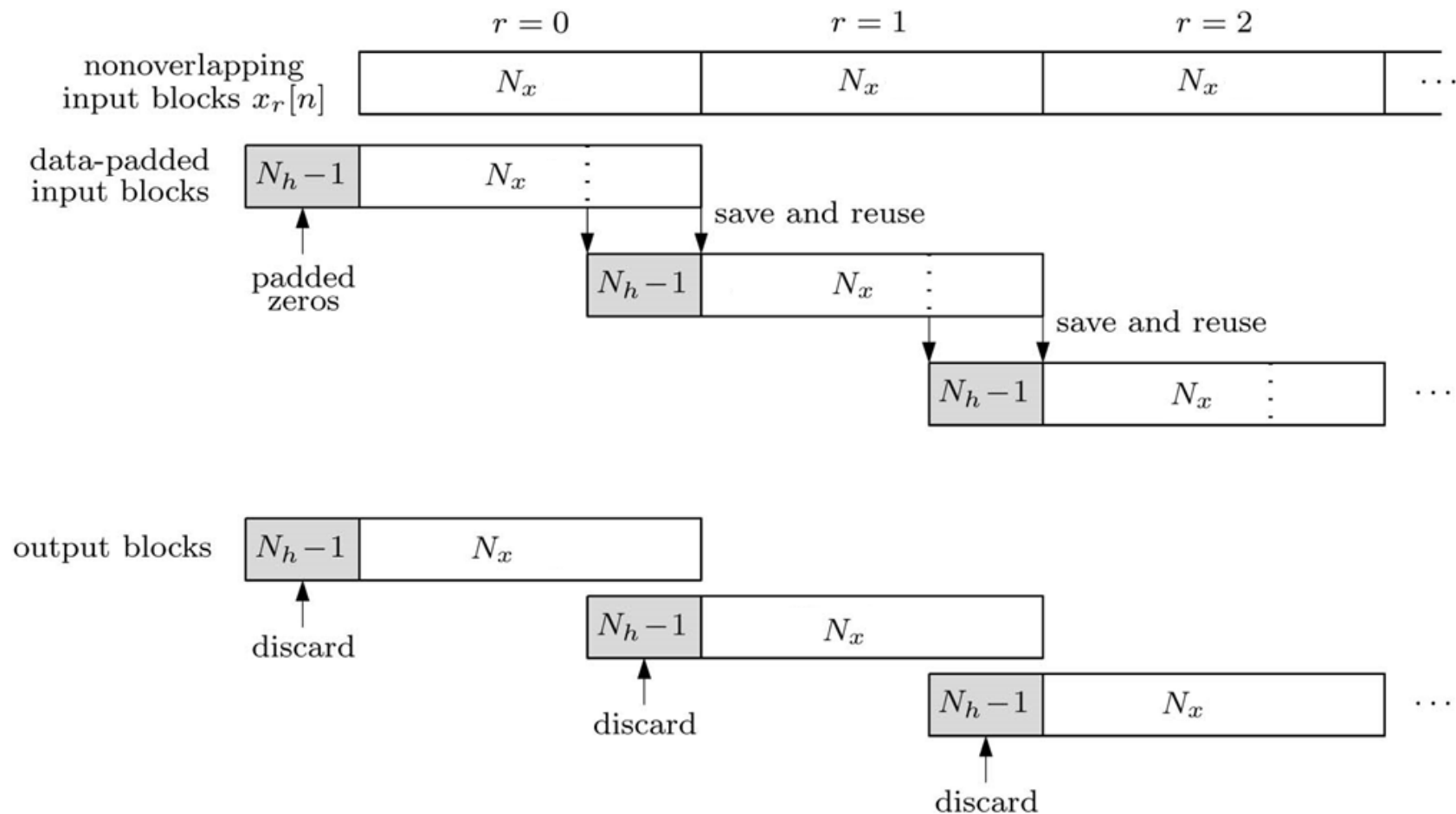
Ans y1 = 2 2 2 2

```
x2pad = [0;1;1;zeros(Nh-1,1)]; X2pad = DN*x2pad; y2 = conj(DN)/N*(X2pad.*Hpad)
```

Ans y2 = 0 2 4 2

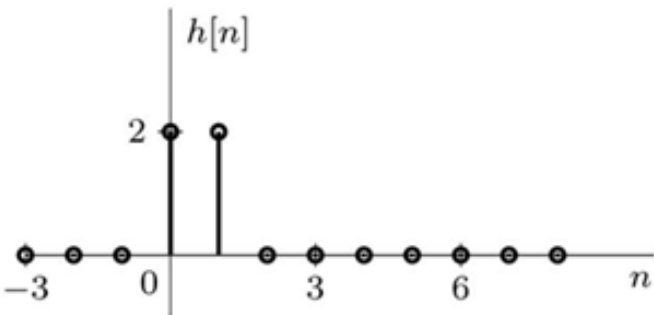
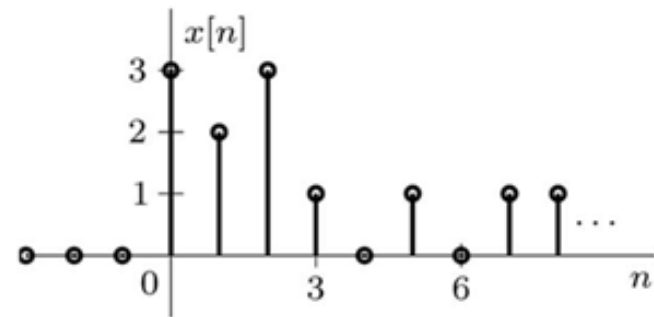
Overlap-and-Save Method

- 1- Section input signal into blocks of length N_x .
- 2- Augment (padded) the beginning of each block with N_h-1 data points from the end of the previous block.
- 3- The first block is augmented with N_h-1 zeros.
- 4- Padd $h[n]$ with N_x-1 zeros to make a length- $N_x + N_h-1$ signal $h_{\text{pad}}[n]$.

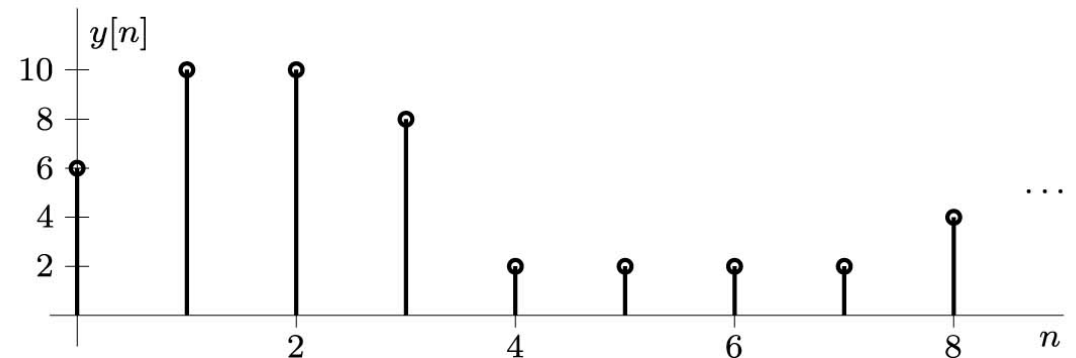
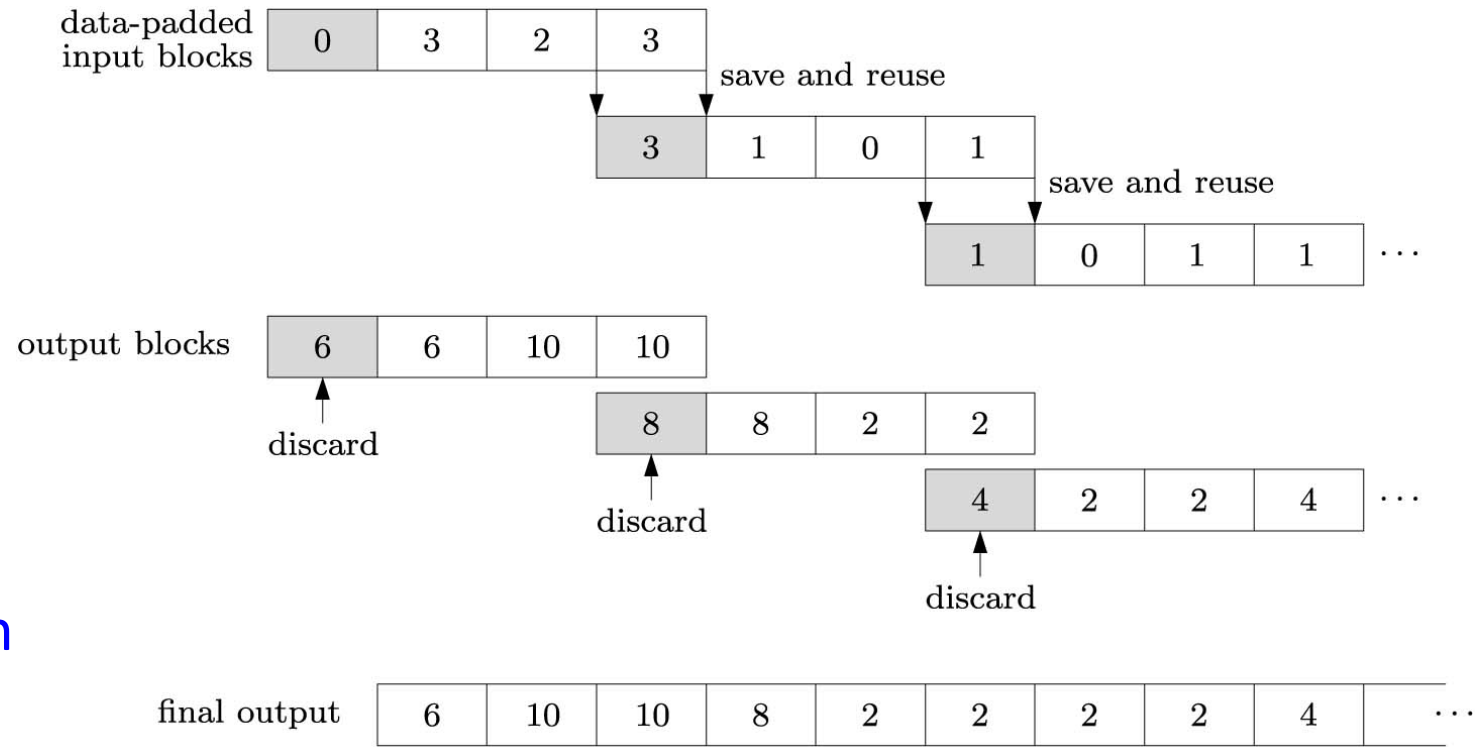


Example

Using the overlap-and-save method with an $N_x = 3$ input block size, find the response $y[n]$ of an LTID system with the $N_h = 2$ finite impulse response $h[n]$ and input $x[n]$ shown in Figure below.



Solution



Continue Example

```
Nx = 4; Nh = 2; N = Nx; Omega0 = 2*pi/N; k = (0:N-1); n = (0:N-1);  
DN = exp(-1j*Omega0*k'*n); x0 = [zeros(Nh-1,1);3;2;3]; hpad = [2;2;zeros(Nx-Nh,1)];  
X0 = DN*x0; Hpad = DN*hpad; y0 = conj(DN)/N*(X0.*Hpad)
```

Ans: y0 = 6 6 10 10

```
x1 = [x0(end-(Nh-1)+1:end);1;0;1]; X1 = DN*x1; y1 = conj(DN)/N*(X1.*Hpad)
```

Ans: y1 = 8 8 2 2

```
x2 = [x1(end-(Nh-1)+1:end);0;1;1]; X2 = DN*x2; y2 = conj(DN)/N*(X2.*Hpad)
```

Ans: y2 = 4 2 2 4

The Fast Fourier Transform (FFT)

FFT is a computational algorithm developed by in 1965 by Tukey and Cooley to reduce the complexity of calculating the DFT from $O(N^2)$ to $O(N \log_2 N)$. The computational saving are substantial for large N . The FFT algorithm is largely what has made the Fourier transform accessible for digital signal processing.

DFT of size $N= 1024$ requires on order of approximately one million operation and FFT reduce it to on order of approximately 10 thousand operation. A reduction of 100 fold. The values obtained by the FFT are identical to those obtained by the DFT.

There are many variation of the FFT algorithm but the two main variation are:

- 1- Decimation in Time (DIT)
- 2- Decimation in Frequency (DIF)

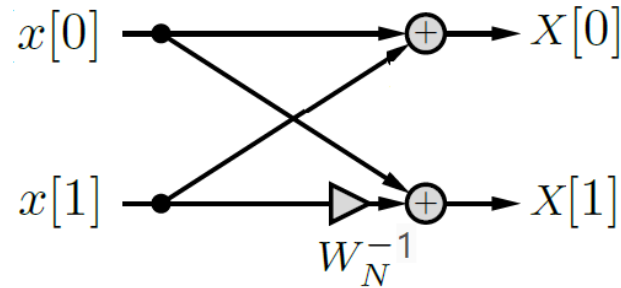
Decimation-in-Time Algorithm of FFT

The DFT of N points:

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j\Omega_0 kn} = \sum_{n=0}^{N-1} x[n] W_N^{-nk} \quad \begin{array}{l} 0 \leq k \leq N-1; \\ \Omega_0 = 2\pi/N; \quad W_N = e^{-j\Omega_0} \end{array}$$

The DFT of 2 points:

$$\begin{aligned} X[0] &= x[0] + x[1] \\ X[1] &= x[0] + W_N^{-1} x[1] \end{aligned}$$



Decimation-in-Time Algorithm of FFT

